



photosynthesis and respiration

Name: _____

Class: _____

Date: _____

Time:

Marks: **68 marks**

Comments:

1.

This question is about photosynthesis.

(a) Complete the word equation for photosynthesis.

_____ + _____ → _____ + oxygen

(2)

(b) Describe how energy for the photosynthesis reaction is gained by plants.

(2)

Students investigated the effect of temperature on the rate of photosynthesis.

The students shone light from a lamp onto pondweed and measured the volume of oxygen produced per hour.

The table below shows the results.

Temperature in °C	Rate of photosynthesis in cm ³ /hour			
	Test 1	Test 2	Test 3	Mean
20	18.5	19.3	19.5	X
25	32.6	34.1	32.9	33.2
30	41.9	45.2	44.9	44.0
35	38.6	39.8	44.0	40.8
40	23.1	20.5	22.4	22.0
45	1.9	14.2	2.2	2.1

(c) Calculate mean value X.

X = _____ cm³/hour

(2)

The students identified one anomalous result in the table above.

(d) Draw a ring around the anomalous result in the table above.

(1)

(e) Suggest **one** possible cause of the anomalous result.

(1)

(f) How did the students deal with the anomalous result?

(1)

(g) Give **one** factor the students should have kept constant in this investigation.

(1)

The table above is repeated below.

Temperature in °C	Rate of photosynthesis in cm ³ /hour			
	Test 1	Test 2	Test 3	Mean
20	18.5	19.3	19.5	X
25	32.6	34.1	32.9	33.2
30	41.9	45.2	44.9	44.0
35	38.6	39.8	44.0	40.8
40	23.1	20.5	22.4	22.0
45	1.9	14.2	2.2	2.1

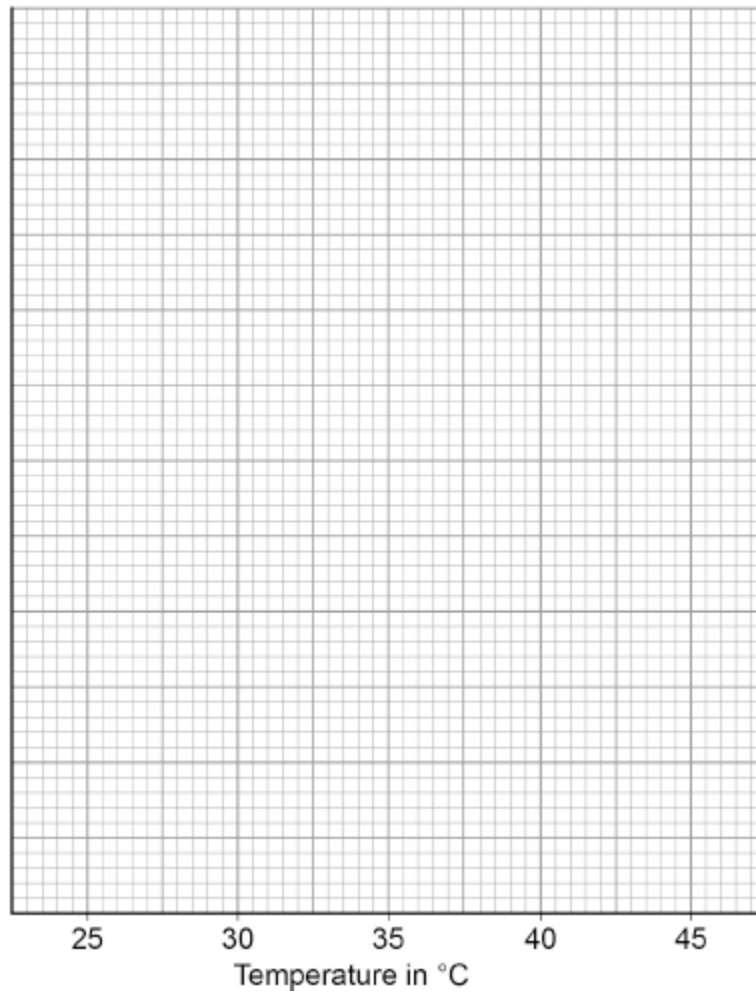
(h) Why did the rate of photosynthesis decrease from 35 °C to 45 °C?

(1)

(i) Complete the graph below using data from the table above.

You should:

- label the y-axis
- use a suitable scale for the y-axis
- plot the mean data from the table above for temperatures from 25 °C to 45 °C
- draw a line of best fit.



(5)

(Total 16 marks)

2.

Green plants can make glucose.

(a) Plants need energy to make glucose.

How do plants get this energy?

(2)

(b) Plants can use the glucose they have made to supply them with energy.

Give **four** other ways in which plants use the glucose they have made.

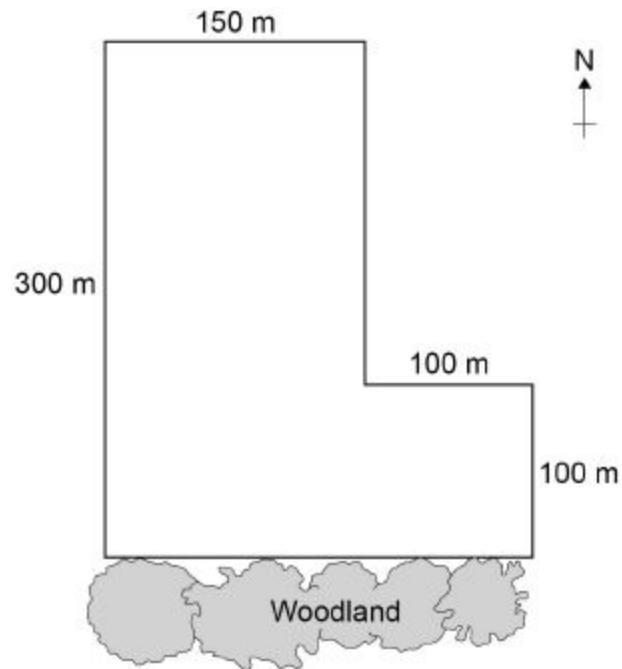
(4)

(Total 6 marks)

3.

Some students investigated the size of a population of dandelion plants in a field.

The diagram below shows the field.



The students:

- placed a 1 m × 1 m square quadrat at 10 random positions in the field
- counted the number of dandelion plants in each quadrat.

The table below shows the students' results.

Quadrat number	Number of dandelion plants
1	6
2	9
3	5
4	8
5	0
6	10
7	2
8	1
9	8
10	11

(a) Why did the students place the quadrats at random positions?

(1)

- (b) Estimate the total number of dandelion plants in the field.

Calculate your answer using information from the diagram and the table above.

Give your answer in standard form.

Total number of dandelion plants = _____

(5)

(d) Light is an environmental factor that affects the growth of dandelion plants.

Give **two** other environmental factors that affect the growth of dandelion plants.

1. _____

2. _____

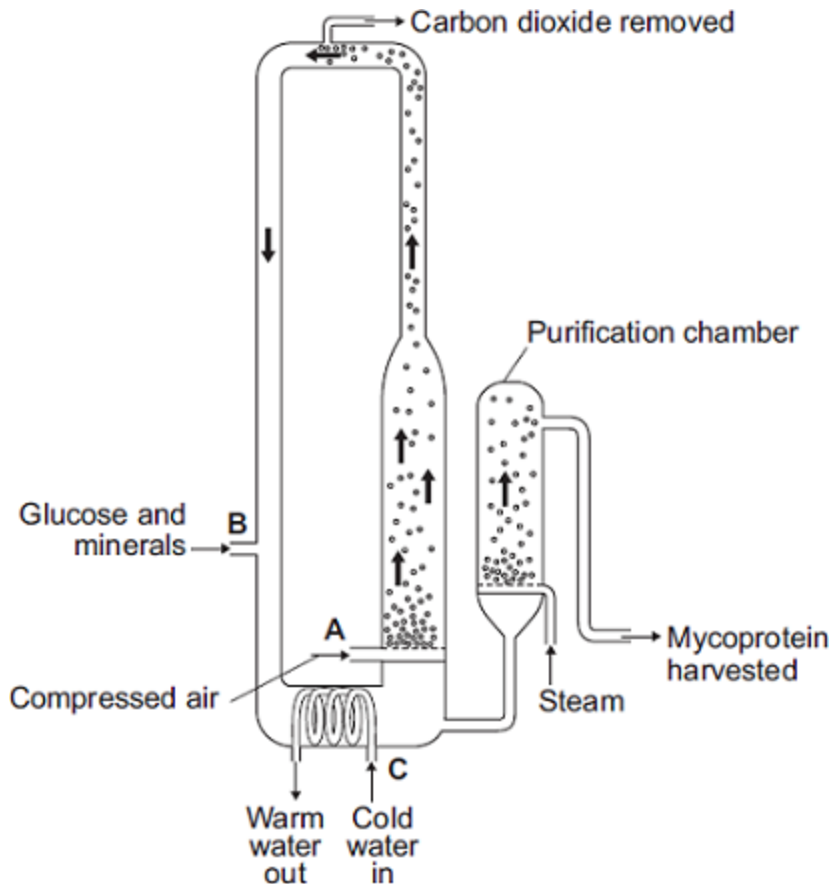
(2)

(Total 14 marks)

4.

The diagram shows a fermenter. This fermenter is used for growing the fungus *Fusarium*.

Fusarium is used to make mycoprotein.



(a) Bubbles of air enter the fermenter at **A**.

Give **two** functions of the air bubbles.

1. _____

2. _____

(2)

- (b) Why is glucose added to the fermenter?

(1)

- (c) The fermenter is prevented from overheating by the cold water flowing in through the heat exchanger coils at **C**.

Name the process that causes the fermenter to heat up.

(1)

- (d) It is important to prevent microorganisms other than *Fusarium* growing in the fermenter.

- (i) Why is this important?

(1)

- (ii) Suggest **one** way in which contamination of the fermenter by microorganisms could be prevented.

(1)

- (e) Human cells cannot make some of the amino acids which we need. We must obtain these amino acids from our diet.

The table shows the amounts of four of these amino acids present in mycoprotein, in beef and in wheat.

Name of amino acid	Amount of amino acid per 100 g in mg			Daily amount needed by a 70 kg human in mg
	Mycoprotein	Beef	Wheat	
Lysine	910	1600	300	840
Methionine	230	500	220	910
Phenylalanine	540	760	680	980
Threonine	610	840	370	490

A diet book states that mycoprotein is the best source of amino acids for the human diet.

Evaluate this statement.

Remember to include a conclusion in your evaluation.

(4)

(Total 10 marks)

5.

(a) Explain, as fully as you can, why respiration has to take place more rapidly during exercise.

(2)

- (b) During exercise the process of respiration produces excess heat. Explain how the body prevents this heat from causing a rise in the core (deep) body temperature.

(4)

(Total 6 marks)

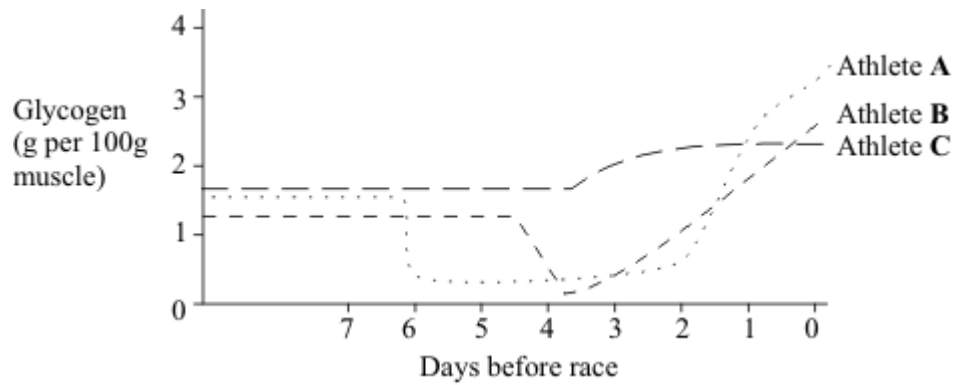
6.

Marathon runners are recommended to have a high carbohydrate diet prior to a race. Three athletes tried out three dietary regimes prior to a marathon race.

These three dietary regimes were as follows.

Athlete A	Up to 7 days before the race	-	Normal mixed diet
	7 days before the race	-	Prolonged extreme physical activity
	6-3 days before the race	-	Protein and fat diet; no carbohydrate
	2 and 1 days before the race	-	Large carbohydrate intake
Athlete B	Up to 5 days before race	-	Normal mixed diet
	5 days before the race	-	Prolonged extreme physical activity
	4-1 days before the race	-	Large carbohydrate intake
Athlete C	Up to 4 days before the race	-	Normal mixed diet
	4-1 days before the race	-	Large carbohydrate intake

The graph below shows the effect of each of these dietary regimes on glycogen levels in the athletes' muscles



- (a) (i) What is the immediate effect of extreme physical activity on the glycogen content of muscles?

(1)

- (ii) Describe how this effect occurs.

(3)

- (b) (i) Evaluate the three regimes as preparation for a marathon race.

(3)

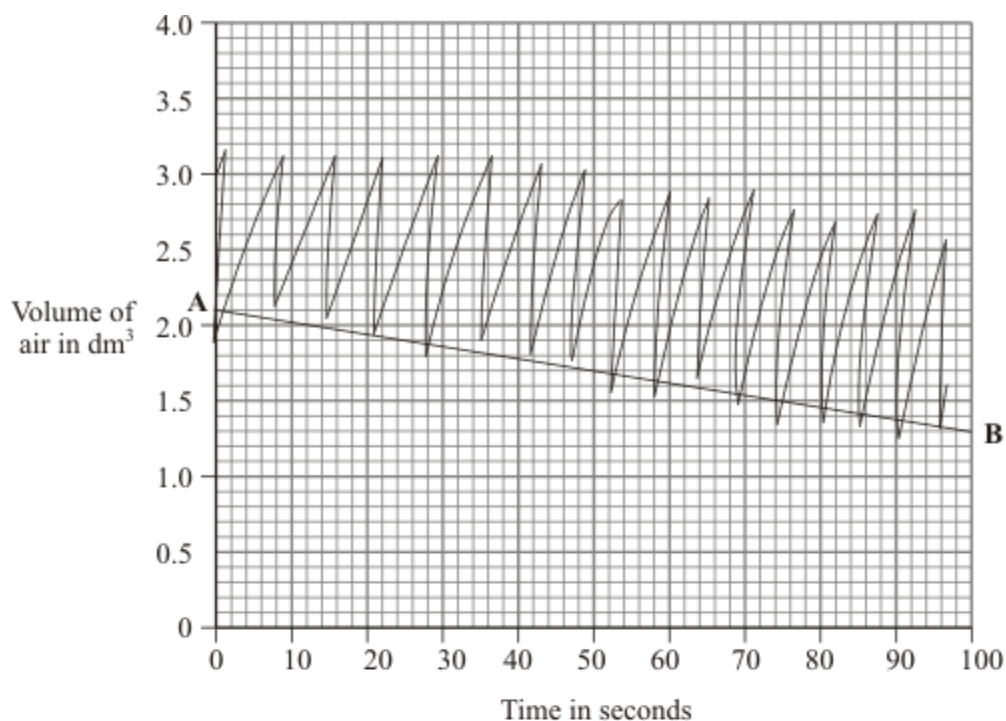
- (ii) Suggest a possible explanation for the different effects of the three regimes.

(2)

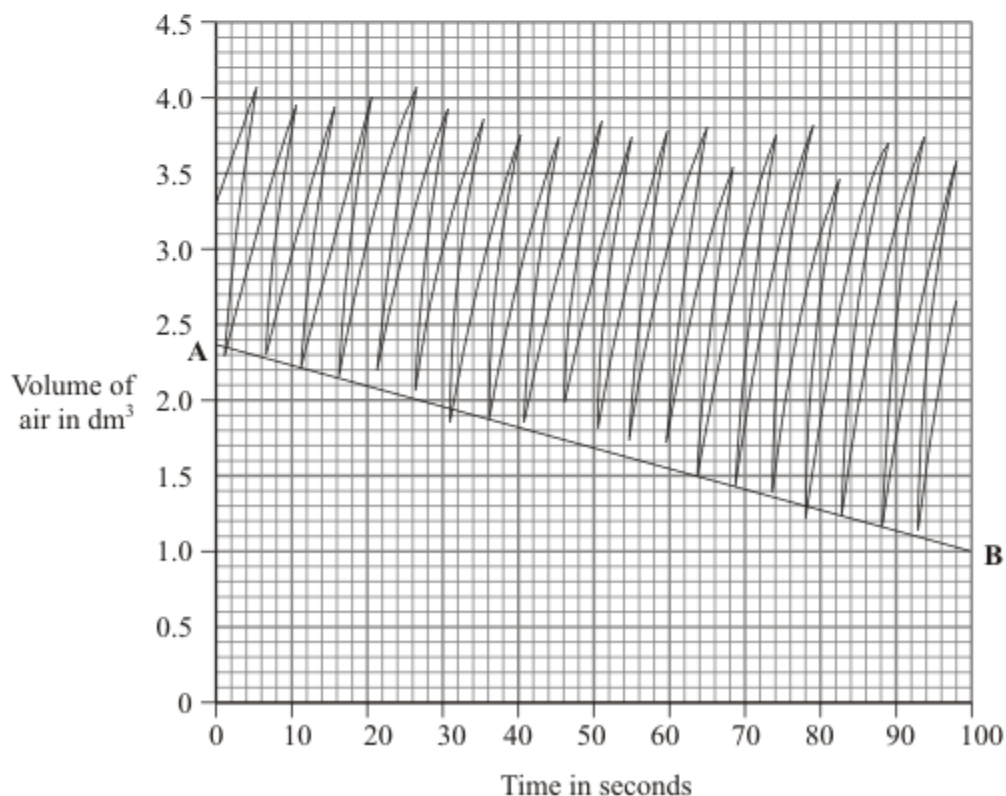
(Total 9 marks)

7.

A student's breathing was monitored before and after vigorous exercise. The student breathed in and out through a special apparatus. The graphs show the changes in the volume of air inside the apparatus. Each time the student breathed in, the line on the graph dropped. Each time the student breathed out, the line went up.



Before exercise



After exercise

- (a) How many times did the student breathe in per minute:

before exercise; _____

after exercise? _____

(1)

- (b) On each graph, the line **A – B** shows how much oxygen was used. The rate of oxygen use before exercise was 0.5 dm^3 per minute. Calculate the rate of oxygen use after exercise.

Rate of oxygen use after exercise = _____ dm^3 per minute

(2)

- (c) The breathing rate and the amount of oxygen used were still higher after exercise, even though the student sat down to rest. Why were they still higher?

(4)

(Total 7 marks)

Mark schemes

1.

- (a) *before arrow*
carbon dioxide **and** water

allow correct chemical symbols
ignore any attempt at balancing equation
ignore light / chlorophyll
either order

1

after arrow

glucose

ignore sugar / carbohydrate
*do **not** accept starch*

1

- (b) light

ignore description of subsequent parts of the photosynthesis reaction
allow sunlight
ignore sun

1

(light) is captured / trapped / absorbed by chlorophyll / chloroplasts
allow (light) is used by chlorophyll / chloroplasts

1

- (c)

$$\frac{(18.5 + 19.3 + 19.5)}{3}$$

or

$$\frac{57.3}{3}$$

1

19.1 (cm³/hour)

allow an answer correctly calculated using only two correct values

1

- (d) a ring around 14.2

allow clear indication of correct result

1

(e) any **one** from:

- scale / value was misread
ignore human error
ignore references to counting bubbles or time
allow measurement error
- there was air / oxygen in the syringe / measuring cylinder / apparatus
- the lamp / light was moved
allow light intensity changed ignore different bulb / lamp unqualified
- temperature changed
- had different mass / length of pondweed
- pondweed had not acclimatised

1

(f) did not use it in calculation (of mean)

1

(g) any **one** from:

- light (intensity)
*do **not** accept temperature*
ignore time
allow distance / power / colour of lamp / light
- carbon dioxide (concentration)
- pondweed size / amount
- pondweed species
allow same (piece of) pondweed

1

(h) enzyme(s) lose the shape of the active site

allow enzyme(s) (start to) denature
allow enzyme(s) destroyed / damaged
*do **not** accept enzyme(s) killed*

1

(i) y-axis labelled '(rate of) photosynthesis in cm³/hour'

1

suitable scale on y-axis

must take up half or more of grid provided

1

all points plotted to within $\pm \frac{1}{2}$ small square

allow 3 or 4 correct plots for 1 mark

ignore any attempt to plot a point at 20 °C

2

correct curved line of best fit

ignore line joined point to point with straight lines

ignore extrapolation

1

[16]

2.

(a) light is trapped / absorbed / used

extra answers cancel mark

ignore solar / sunshine

1

by chlorophyll / chloroplasts

if no other marks awarded, allow 1 mark for photosynthesis /

equation for photosynthesis

1

(b) (to make) starch (for storage)

ignore 'for growth' unqualified

ignore respiration

1

(to make) fat / oil (for storage)

1

(to make) amino acids / proteins / enzymes

1

(to make) cellulose / cell walls

allow for active transport

allow any other correct, named organic substances (eg DNA / ATP / chlorophyll / hormone)

*if no named examples, allow 'to make **named** cell structures' for max. 1 mark*

1

[6]

3.	(a) there is an uneven distribution of dandelions or (more) representative / valid or avoid bias or more accurate / precise mean <i>ignore accurate / precise unqualified</i> <i>ignore repeatability / reproducibility / reliability / fair test</i>	1
	(b) (correct mean per m ² ⇒) 6 or 6.0	1
	(correct field area ⇒) 55 000 (m ²)	1
	mean × area – e.g. 6(.0) × 55 000 <i>allow incorrect calculated values for mean and / or field area</i>	1
	330 000 <i>allow correct calculation from previous calculation</i>	1
	3.3×10^5 <i>allow calculated value in standard form</i> <i>an answer of 3.3×10^5 scores 5 marks</i> <i>an answer of 330 000 scores 4 marks</i>	1
	(c) Level 3: The method would lead to the production of a valid outcome. All key steps are identified and logically sequenced.	5–6
	Level 2: The method would not necessarily lead to a valid outcome. Most steps are identified, but the method is not fully logically sequenced.	3–4
	Level 1: The method would not lead to a valid outcome. Some relevant steps are identified, but links are not made clear.	1–2
	No relevant content	0

Indicative content

- placing of quadrat
- large number of quadrats used
- how randomness achieved – e.g. table of random numbers **or** random number button on calculator **or** along transect
- quadrats placed at coordinates **or** regular intervals along transect
- in each of two areas of different light intensities **or** transect running through areas of different light intensity
- for each quadrat count number of dandelions
- for each quadrat measure light intensity
- compare data from different light intensity

to access **level 3** the key ideas of using a large number of quadrats randomly, or along a transect, and counting the number of dandelions in areas of differing light intensity need to be given to produce a valid outcome

(d) any **two** from:

- temperature
allow heat
- water
allow moisture / rain
- (soil) pH
allow acidity
- minerals / ions
*allow e.g. magnesium ions **or** nitrate*
allow salts / nutrients
- winds
- herbivores
allow trampling
ignore carbon dioxide
ignore space
ignore competition unqualified
*do **not** accept oxygen*

2

[14]

4.

(a) circulating / mixing / described **or** temperature maintenance

1

supply oxygen

or for aerobic conditions

or for faster respiration

*do **not** allow oxygen for anaerobic respiration*

1

- (b) energy supply / fuel / use in respiration
*do **not** allow just food / growth*
ignore reference to aerobic / anaerobic

or material for growth / to make mycoprotein

1

- (c) respiration

allow exothermic reaction
allow catabolism
ignore metabolism
ignore aerobic / anaerobic

1

- (d) (i) any **one** from:

- compete (with *Fusarium*) for food / oxygen **or** reduce yield of *Fusarium*
- make toxic waste products or they might cause disease / pathogenic **or** harmful to people / to *Fusarium*
*do **not** allow harmful unqualified*

1

- (ii) steam / heat treat / sterilise fermenter (before use)
***not** just clean*

or

steam / heat treat / sterilise
glucose / minerals / nutrients / water (before use)

or

filter / sterilise air intake

or

check there are no leaks

*allow sterilisation unqualified **not** just use pure glucose*

1

- (e) any **three** from:

- beef is best or beef is better than mycoprotein
- mycoprotein mainly better than wheat
- more phenylalanine in wheat than in mycoprotein
allow equivalent numerical statements
- but no information given on other amino acids / costs / foods

3

overall conclusion:

statement is incorrect because

either

it would be the best source for vegetarians

or

for given amino acids, beef is the best source

or

three foods provide insufficient data to draw a valid conclusion

1

[10]

5.

- (a) more energy needed,
for increased muscular activity

for 1 mark each

2

- (b) increased sweat production,
evaporation of sweat cools body,
vasodilation OWTTE,
more heat loss (by radiation)

for 1 mark each

4

[6]

6.

- (a) (i) reduced sharply

for 1 mark

1

- (ii) converted to glucose which is respired to produce energy

(allow answers in terms of glucagon)

gains 3 marks

3

- (b) (i) athlete As was most effective
since resulted in highest muscle glycogen level on day of race
for energy release during race

for 1 mark each

3

- (ii) e.g. excess carbohydrate stored as glycogen rather than fat in short term
particularly if glycogen stores depleted

for 1 mark each

2

[9]

7.

- (a) (before exercise) – 9 to 11 **and** (after exercise) – 12 **or** 13

both correct

1

(b) 0.75 to 0.90

ignore working or lack of working

eg. $2.35 - 1.55$ **or** $\frac{(2.35 - 1.0) \times 60}{100}$ **or other suitable figures**
for **1** mark

2

(c) any **four** from:

still need to remove extra carbon dioxide

still need to remove heat / to cool

(some) anaerobic respiration (in exercise)

lactic acid made (in exercise)

oxygen needed to break down lactic acid **or** suitable reference to oxygen debt

lactic acid broken down to CO₂ and water **or** lactic acid changed into glucose

4

[7]

Examiner reports

- 1.** The equation for photosynthesis had been generally well learnt, with two-thirds of students scoring at least one mark. (b) was not answered well, with the reference to 'energy' leading some students to write confused accounts about respiration. Nearly two-thirds of students were able to identify the anomalous result in (d). In (e), reference to 'human error' on its own was insufficient.

The concept of denaturation in (h) was not acknowledged by most students, as they were unable to make the link between photosynthesis and enzymes. Students are advised to read and follow all of the instructions given when asked to draw a graph, as in (i), many students missed key tasks despite them being listed clearly. Students need to understand how to draw a line/curve of best fit when the plots do not fall on a straight line and students generally could be more accurate when plotting points, particularly those that are not whole numbers. A few students plotted the three individual repeats for each temperature rather than the mean, which meant that no plotting marks could be awarded.

- 2.** This question was about the production of glucose by green plants and its possible subsequent use by the plants.

- (a) More than three quarters of students were able to make at least one point about how plants obtained the energy required for the production of glucose. This was generally for stating that *light* was the source or that *photosynthesis* was the process. Fewer mentioned the need for chlorophyll or chloroplasts to trap the light energy.
- (b) This question differentiated well across the ability range. It required specific knowledge of sections 2.3.1 e and f in the Specification. Most knew that glucose could be converted to starch, but few could mention amino acids / protein, fats / oils, or cellulose / cell walls in addition. Many mentioned 'respiration', despite this being precluded by the wording of the question. Many wrote at length although a simple list of four points would have sufficed – time would have been better spent on thinking than on writing, especially if the writing simply made vague reference to 'growth', 'repair' and 'reproduction' with no specific mention of how glucose could contribute to these processes.

- 3.** (a) 57% of students understood that the purpose of placing quadrats at random was to:
- allow for the uneven distribution of the dandelion plants
 - get a more representative or valid result
 - avoid bias
 - obtain a more accurate or precise mean value from the 10 samples.

Many others stated that it was to make the results accurate, precise, reproducible, repeatable or fair and such statements were ignored by examiners.

- (b) Calculation of the dandelion population in the L-shaped field was a 5-step process, with each step worth a mark:
1. calculation of the mean number of dandelions in 1 m² from the 10 results in the table (6 / m²)
 2. calculation of the area of the field (55 000 m²)
 3. multiplying the mean by the area (6 × 55 000)
 4. getting the correct answer from step 3 (330 000)
 5. converting the answer from step 4 to a number in standard form (3.3 × 10⁵).

Errors made at each step were allowed for in subsequent steps to avoid excessive loss of marks when students clearly understood the method they had to employ. Many students set out their working in a somewhat disorganised way which was difficult for examiners to follow.

44% of students achieved full marks, with 29% gaining four out of the five marks, primarily because they forgot to convert their final answer to a number in standard form.

- (c) Many students produced excellent answers to this question. But a large proportion of the remainder did not achieve any marks at all as they chose an inappropriate investigation: sowing dandelion seeds in pots and seeing how well they grew with lights of different intensities shining on them.

Higher-attaining students described how they would run a line transect from the shady woodland across the field, placing the quadrat at regular intervals along this and counting the dandelion plants in each quadrat. They anticipated that the density of dandelions would increase further away from the woodland.

The highest-attaining students stated that a light reading should be taken with a light meter within each quadrat, that the transect should be repeated at several other similar locations, means being calculated for set distances from the woodland, and that a graph should be plotted of the number of dandelions against light intensity, hoping to find a positive correlation. A suitable alternative to this method was to place a large number of random quadrats in a shady area and in a brightly-lit area and determining the mean number of dandelions for each area which could then be compared.

Answers were marked using a level of response mark scheme with:

- level 3 (5–6 marks) being awarded if the method described would give a valid outcome and the key steps were logically sequenced
- level 2 (3–4 marks) for a sensible method which would not necessarily produce a valid outcome as one or two key steps had been omitted.
- level 1 (1–2 marks) giving an answer that would definitely not give a valid outcome, although some relevant material had been included.

Overall, 33% of students achieved level 3.

- (d) Suitable suggestions of two other environmental factors that might have determined how well the dandelions grew included water, mineral ions and the pH of the soil, as well as biotic factors such as herbivores or trampling. 97% of students were able to give at least one factor with 44% managing to provide two.

Factors considered inappropriate included the concentration of carbon dioxide or oxygen in the air, as they were unlikely to be limiting in an open field, but these were often stated by students.

4.

- (a) Around two thirds of students were able to think of at least one use of the bubbles of air introduced near the base of the fermenter, but only a quarter were able to suggest both *supplying oxygen* and *circulating materials* around the fermenter. Some thought it was 'to remove carbon dioxide', forgetting that aerobic respiration would actually generate carbon dioxide.
- (b) Around two thirds got this right. Some gave inadequate answers such as the glucose merely being 'food' or being 'for growth'; if they had stated that the glucose was potentially a supply of *building material* for growth, then this would have scored the mark.
- (c) Just over half of students appreciated that cooling of the fermenter was necessary due to heat having been generated by the process of *respiration*.
- (d) Approximately one third of students were able to explain why it was important to exclude microorganisms other than the *Fusarium* from the fermenter. Correct answers included reference to avoidance of competition with the *Fusarium* for resources such as glucose and oxygen, or the production of toxic products by the contaminating microorganisms. Conversely, over two thirds knew that sterilisation of apparatus and materials before use, or filtering the air intake, or sealing up any leaks would help to prevent contamination with microorganisms.
- (e) Most students recognised, from the given table of data, that beef was the best source of the four stated amino acids. However, far too many concentrated on making comparisons between amino acid content and the information on daily amounts needed, assuming that a person could only consume 100 grams of each product (mycoprotein, beef or wheat). Therefore comparisons with the human daily requirement for each amino acid were invalid. Also a lot of students misinterpreted the word 'diet' and immediately talked in terms of which food would be most appropriate for weight loss. While most students were able to extract at least one valid point from the data, very few managed to score all four marks. The question differentiated well across the ability range. Many also went beyond the remit of the actual question and considered the overall quality of the different foods, e.g. beef being high in fat or growing wheat rather than beef being better in terms of reducing environmental impact. Some also expressed their disapproval of eating manufactured protein and stated they did not like the taste either!

5.

- (a) There was a great deal of confusion between respiration and breathing, indeed to answer in terms of "getting more oxygen to the muscles" was almost the norm.
- (b) Incorrect phrases abounded, ruining the answers of a majority of candidates: "blood vessels rise to the surface"; "hairs lie flat to keep the heat in"; "sweating opens pores to let the heat out" and "heat evaporates from the body" were seen far too often by all examiners.

6.

Many candidates did not seem to understand the meaning of 'evaluate' and consequently received only one mark – for stating that A was the most effective diet – losing those for giving reasons why. Very few reasonable suggestions were seen – it was sufficient to say that emptying the muscle glycogen stores a few days before the race meant that when a high carbohydrate was taken two days before the race most of the excess carbohydrate went directly into muscle storage rather than into fat stores.

7.

Some candidates mistakenly believe that there are one hundred seconds in a minute. This led to obvious errors in both parts (a) and (b). Responses to part (c) were often good and even the least able candidates referred to lactic acid and oxygen debt.